Qn 2	Answer	Marks
(a) (i)	For circular motion, centripetal force is provided by the gravitational force, $$\frac{GM_J m}{R^2} = mR\omega^2$$ $$G\frac{M_J m}{R^2} = mR\left(\frac{2\pi}{T}\right)^2 \Rightarrow T = \sqrt{\frac{4\pi^2 R^3}{GM_J}}$$	**M1** **A1**
(a) (ii)	Since $T^2 \propto R^3$, $$\left(\frac{T_{Th}}{T_{Am}}\right)^2 = \left(\frac{R_{Th}}{R_{Am}}\right)^3 \Rightarrow \left(\frac{0.676}{T_{Am}}\right)^2 = \left(\frac{3.18}{2.62}\right)^3$$ T_{Am} = 0.506 Earth-days	**M1** **A1**
(b)(i)	$$v = \frac{2\pi R}{T} = \sqrt{\frac{GM_J}{R}}$$ <u>OR</u> Since the orbit is circular, centripetal force is provided by gravitational force, hence $$\frac{GM_J m}{R^2} = \frac{mv^2}{R} \Rightarrow v = \sqrt{\frac{GM_J}{R}}$$ A1	**A1**
(b)(ii)	When the mass of the moon decreases, although the gravitational force and centripetal force required both decreases, the condition for orbit, that the gravitational force is **equal** to the centripetal force, is still true. Hence the moon will stay in orbit.	**B1**
(c)(i)	Read off graph with orbital period equal to one Jupiter-day = 0.417 Earth-days Orbital radius of geostationary orbit is <u>2.30 R_J</u>. (Allow : ± ½ smallest div)	**A1**
(c)(ii)	Able to c<u>ontinuously observe</u> the <u>same area</u> on Jupiter for an extended period of time.	**B1**
	[Total : 8 marks]	